DATE	28-02-2026
TEAM ID	LTVIP2026TMIDS90614
PROJECT NAME	iRevolution: A Data-Driven Exploration of Apple's iPhone Impact in India Using Tableau
MAXIMUM MARKS	4 MARKS

CHAPTER-7

Functional and Performance Testing

7.1 - Functional and Performance Testing:

Functional and Performance Testing are essential stages in project validation to ensure that the developed system operates correctly, efficiently, and reliably. In a data analytics project such as "iRevolution: A Data-Driven Exploration of Apple's iPhone Impact in India Using Tableau," testing ensures that dashboards, calculations, filters, and data connections function accurately and deliver expected results without errors or delays.

1. Functional Testing

1.1 Concept of Functional Testing

Functional testing refers to the process of verifying whether all components of the system perform according to specified requirements. It focuses on validating features, calculations, data accuracy, and user interactions.

In this project, functional testing ensured that:

- Data was correctly imported into Tableau
- Calculated fields produced accurate results

1.2 Functional Testing Activities

The following checks were performed:

- Verification of dataset accuracy after import
- Cross-checking calculated metrics (Growth %, Revenue totals, Market Share)
- Testing filter functionality (Year, Region, Model)

1.3 Importance of Functional Testing

- Ensures correctness of analysis
- Prevents misleading insights
- Improves reliability of dashboards

2. Performance Testing

2.1 Concept of Performance Testing

Performance testing evaluates how efficiently the system performs under expected workload conditions. In Tableau dashboards, performance refers to loading speed, responsiveness, and smooth interaction.

2.2 Performance Testing Activities

In this project, performance was tested by:

- Checking dashboard loading time
- Testing filter response speed
- Evaluating performance with large datasets
- Verifying smooth navigation between story points

2.3 Importance of Performance Testing

- Ensures smooth user experience
- Reduces delays and system crashes
- Improves dashboard efficiency